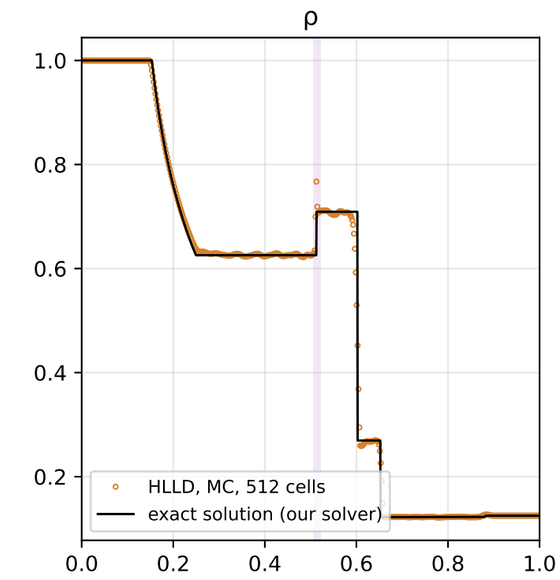

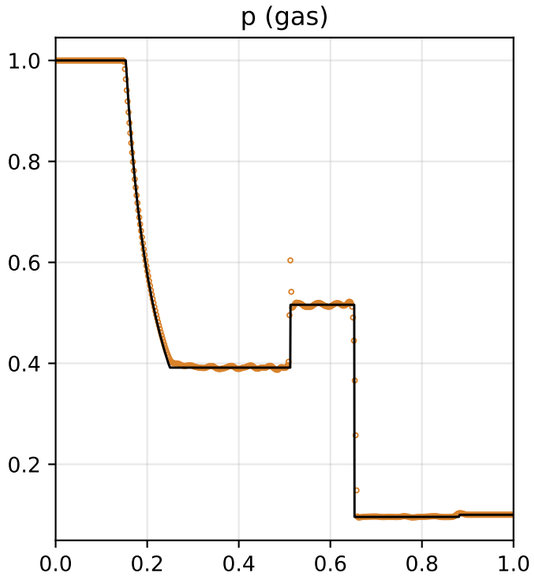

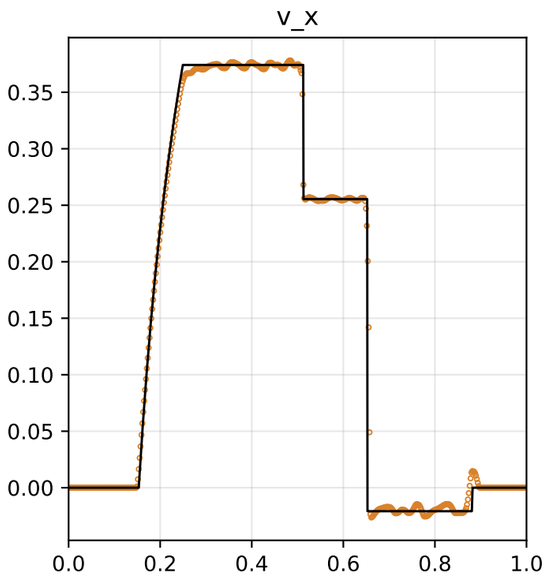

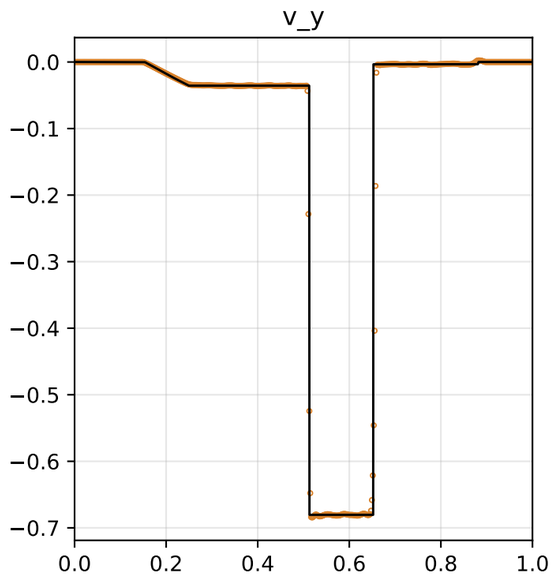

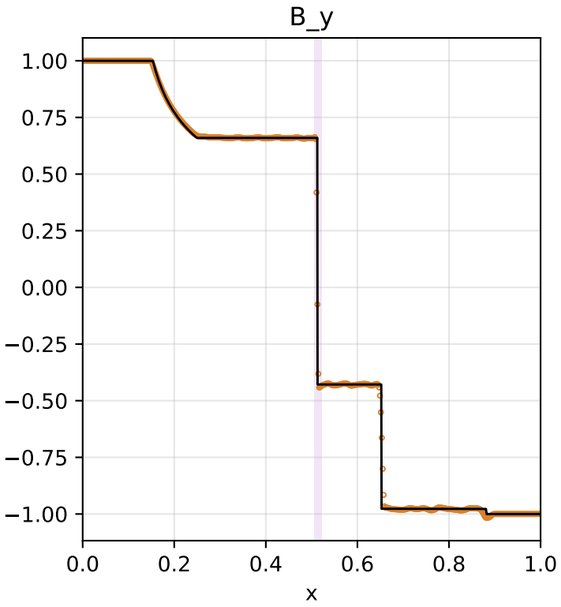

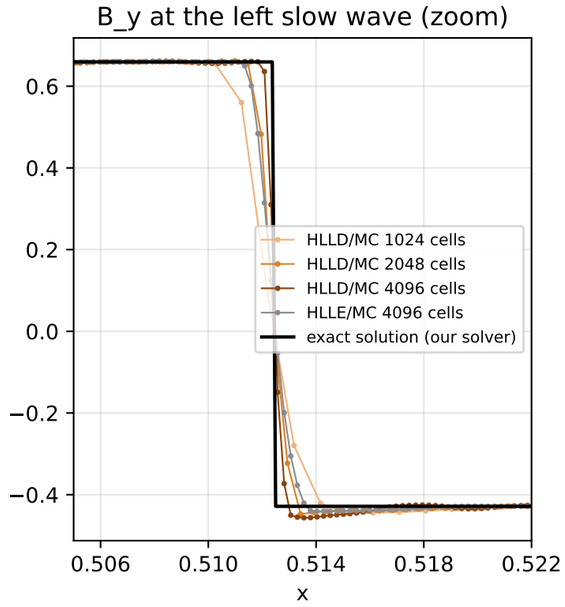

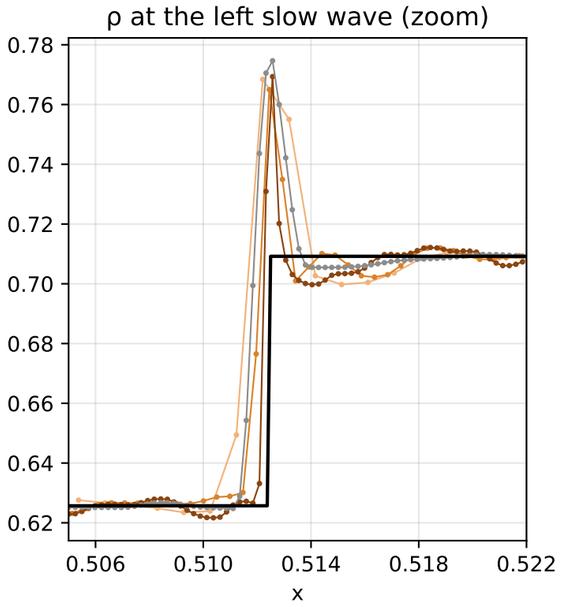

Balsara 1 (relativistic Brio–Wu), Γ = 2, t = 0.4
L: ρ = 1, p = 1, B = (0.5, 1, 0); R: ρ = 0.125, p = 0.1, B = (0.5, −1, 0)

Waves, left to right: fast rarefaction, a slow-family shock that REVERSES B_y (+0.66 → −0.43), contact, slow shock, weak fast rarefaction. Both Alfvén waves carry nothing.

The reversing wave is an intermediate shock: the gas crosses the Alfvén speed through it. Checked: jump conditions to 1e-13, entropy rises across both shocks, the runs converge to it, and it is the only root found from 224 starts (none with a 180° Alfvén rotation).

The spike in ρ, p behind it is the scheme's overshoot, about two cells wide at every resolution.